

Pricing Change Causal Impact Assessment

Executive Decision Summary

Business Question

Should the company keep, rollback, or redesign the recent price increase introduced for a mid-tier product plan?

Context

The price increase was rolled out without randomization. User exposure depended on observable attributes such as device type and acquisition channel, introducing selection bias and making raw performance metrics unreliable.

Methods Used

Difference-in-Differences, Propensity Score Matching, bootstrap confidence intervals, placebo rollout date tests, randomized placebo treatment, and negative control outcome validation.

Key Findings

Conversion Impact (Matched Users): -2.36 percentage points (95% CI: -2.93 to -1.83).

Revenue Impact: Net revenue per exposed user declined. Conversion losses outweighed price uplift.

Validation: All placebo and falsification tests showed no false effects, supporting causal credibility.

Final Recommendation

Rollback or redesign the price increase. After correcting for selection bias and validating results with multiple falsification tests, the price change caused a statistically significant drop in conversion and reduced expected revenue.

Confidence Level

High. Results are consistent across matching-based causal analysis and validated through placebo and negative control tests.